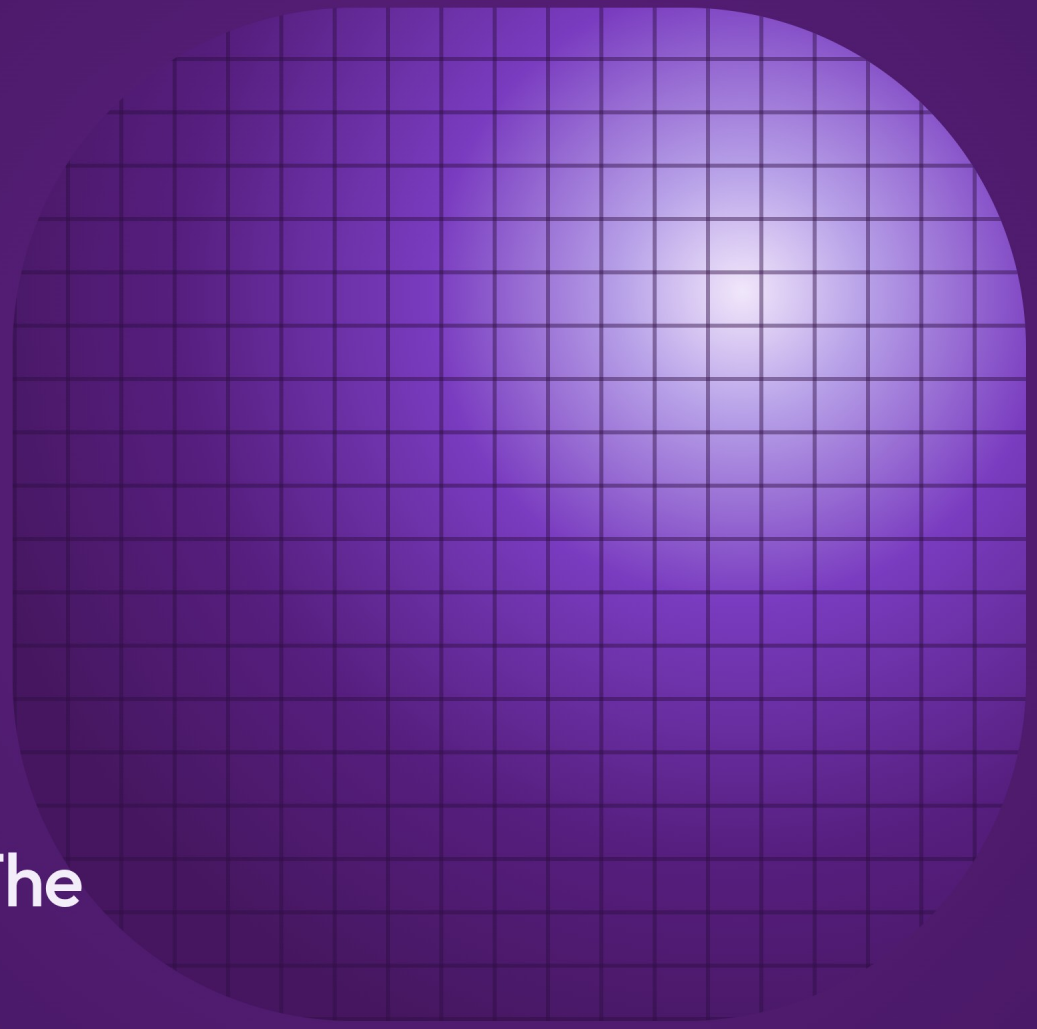
> HACKATHON 2026 // SERENEDB CHALLENGE

Log Insights AI Agent

The right insights for the right people: engineer, team lead, manager.



The database searches & calculates. The agent explains.

Engineer | Team lead | Manager

Celil Durna · Berkan Gürbüz

> THE PROBLEM

Three readers, one set of logs



01 · Engineer

What failed, and what happened just before?

Needs a concrete diagnostic check.



02 · Team lead

What recurs, how widely, and who owns it?

Needs recurrence, spread and ownership.



03 · Manager

What comes first, and how sure are we?

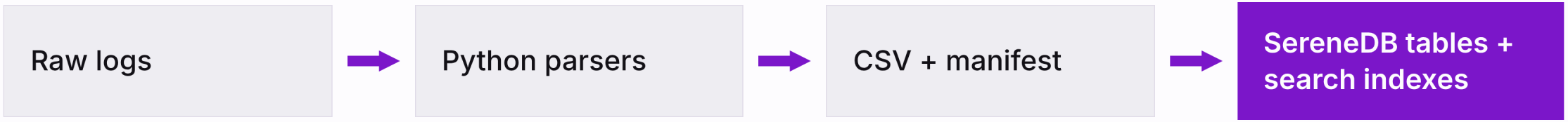
Needs comparable priorities and explicit limits.

SereneDB searches and calculates. The agent explains what the queries returned.

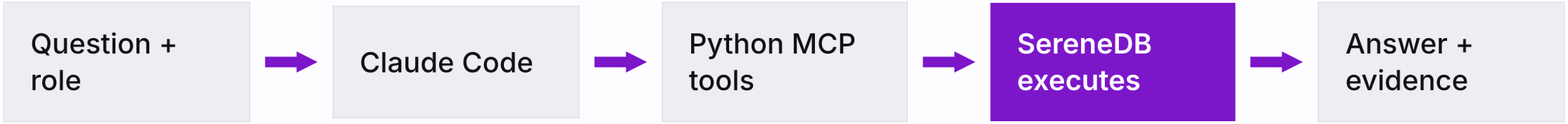
> ARCHITECTURE // LOCAL PIPELINE

Full logs, loaded into a local SereneDB

PREPARE · ONCE



ASK · EVERY QUESTION



4,747,963
BGL log lines

207,820
OpenStack log lines

LOG STORAGE STAYS LOCAL. ONLY QUERY RESULTS GO TO THE MODEL.

> TOOL DESIGN

Six read-only tools, one job each

01

describe_dataset

What is in the data: size, time range, labels.

02

search_logs

Full-text search, word or phrase, with time, label and level filters.

03

nearby_events

Lines before and after an anchor line, on one node or on all.

04

count_by

Grouped counts by label, component, team, rack or node.

05

trend

Counts per minute or per day, for any filter.

06

ping

Checks the connection to the database.

Every answer carries its evidence: reported SQL, parameters, scope, row counts and a truncation flag.

> SEARCH VERIFICATION // EXACT VS BROAD

Search and complete counts agree on APPREAD

5,983

Phrase matches for “failed to read message prefix”

5,983

Complete count of the label APPREAD

350,567

Word search for “ciod”: broader, matches every ciod line

tool call | sql evidence

```
{"dataset":"bgl","text":"failed to read
message prefix","phrase":true}
```

> SQL, SHORTENED

```
SELECT COUNT(*) FROM bgl_logs_message_idx
WHERE message @@ phraseto_tsquery(%
(text)s);

SELECT COUNT(*) FROM bgl_logs WHERE label =
%(label)s;
```

FOR THIS LABEL AND PHRASE, INDEX AND FULL TABLE GIVE THE SAME ANSWER.

> 01 · ENGINEER // 16 JUNE 2005

What failed, and what happened just before?

−0.575 s

chdir failed on another node

−0.540 s

chdir failed on another node

0 s · anchor

APPREAD: failed to read message prefix

+0.05 s…

15 more APPREAD lines on other nodes, same target host

> OBSERVED

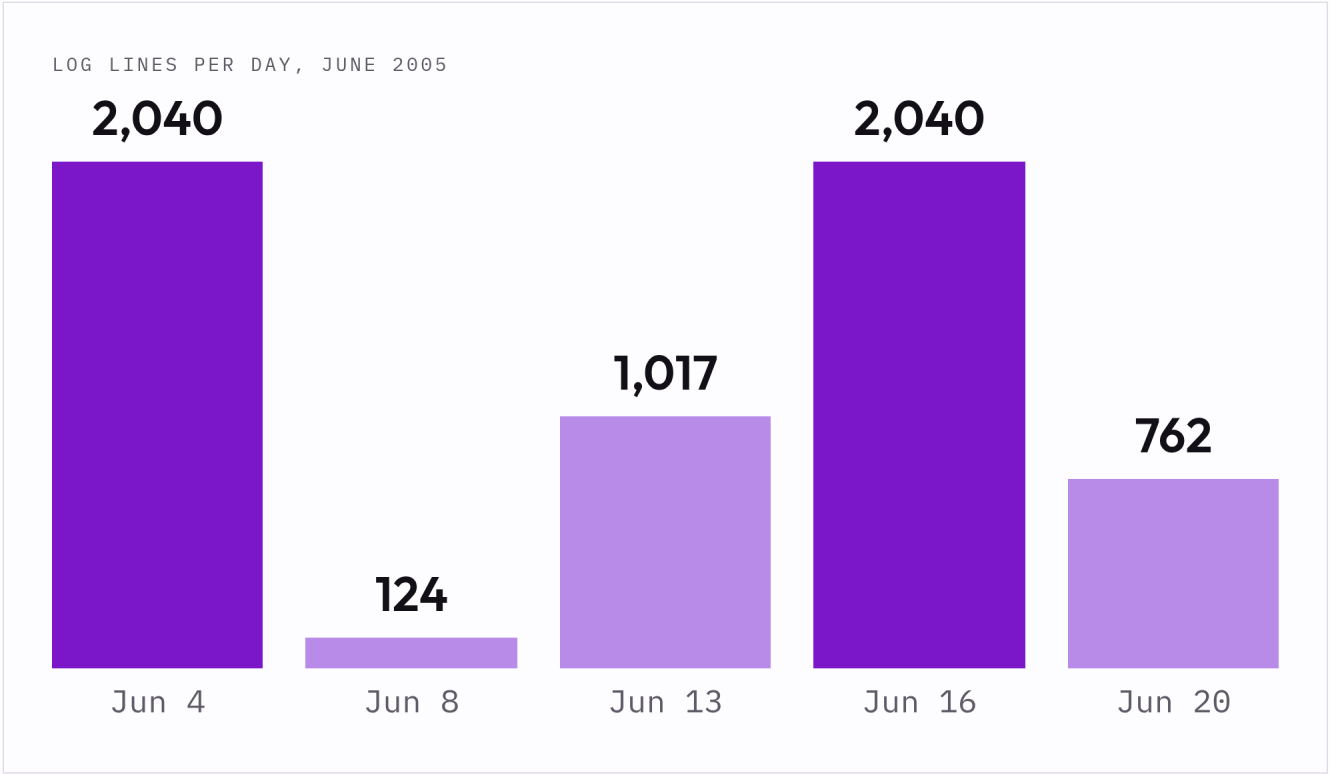
2,040 lines from 510 nodes, 4 lines per node, in minute buckets from 14:50 to 15:03. Same pattern and host in the 20 lines inspected.

> HYPOTHESIS, NOT PROVEN

A job may have failed to start because its working directory was missing. Timing only. The 4 June burst shows the same two chdir failures 0.8 s before its first line.

> 02 · TEAM LEAD

APPREAD returns on five days, two of them equal peaks



5,983
log lines on 5 active days

`APP · Application Runtime Team`
Team mapping is fictional.

> Action: start with 4 and 16 June (2,040 lines and 510 nodes each) and compare their shared failure context.

COUNTS ARE LOG LINES, NOT INCIDENTS.

> 03 · MANAGER

Assign APPSEV and KERNMNTF first

#	PROBLEM	LOG LINES	ACTIVE DAYS	PEAK SHARE	STATUS
01	**APPSEV**	49,651	67	7.0%	Still active
02	**KERNMNTF**	31,531	25	13.8%	Still active
03	**KERNTERM**	23,338	28	17.0%	Ended 2005-11-16

> ONE-OFF REVIEWS

KERNDTLB (99.6% of its lines on 11 June) and KERNSTOR (74.2% on 14 June) are concentrations, not recurring problems.

> CHECKED AGAINST THE RULE

An earlier run labelled APPREAD “limited”. By the rule it is “recurring” (5 active days, 34.1% peak share). Later runs did not repeat it.

RULE: BURST IF ONE DAY HOLDS OVER HALF OF THE LINES; RECURRING IF ACTIVE ON 5+ DAYS. PRIORITY FOLLOWS RECURRENCE, NOT BUSINESS IMPACT.

> TRUST // RIGOR

Verified in layers

01

5/5

trend reference cases, checked against known baselines

02

54/54

corpus validation checks: schemas, timestamps, record counts

03

2

clean-clone runs from scratch, on Linux and on Windows

> Tests: 28 offline tests and 3 targeted DB tests passed; the full DB suite encountered SereneDB crashes.

> Evidence on every answer: reported SQL, parameters, scope, row counts, truncation flags.

CLEAN-CLONE RUNS WERE DONE AT EARLIER COMMITS, NOT AT THE FINAL ONE.

> NEXT STEPS

Recommendations and limits

01 · Engineer

Inspect the other active days and the logs of the remote endpoint.

02 · Team lead

Compare the shared failure context of the two peak days.

03 · Manager

Assign APPSEV and KERNMNTF first; for KERNTERM check change records. OpenStack: 192 periodic-task ERROR lines; recurrence not assessable over the short recording period.

> LIMITS

Research logs, not production monitoring

Proximity is not causation

Team ownership is fictional

Log lines are not independent incidents

No cost or downtime data

Occasional SereneDB crashes (SIGSEGV), cause unknown

> LOG INSIGHTS AI AGENT

One database for search and analytics. Evidence behind every answer.



Engineer: a concrete diagnostic check

Team lead: recurrence, spread, ownership

Manager: comparable priorities, explicit limits

github.com/celil-durna/log-insights-ai-agent